VIJAYBHOOMI UNIVERSITY

# RESEARCH INCUBATION HANDBOOK

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# 1.0 Introduction

The Research Incubation (RI) program at Vijaybhoomi University aims to equip students with critical thinking and problem-solving skills. This program will facilitate students to develop research-oriented thinking which will enable them to thrive and innovate in today's fast-changing business and technical environment.

This Research Incubation program applies only to UG Programs and is offered in every term. For a 3-year UG program, students can register for an RI course after completing one and a half years and 60 credits. For the 4 and 5-year UG program, students need to register for RI after completing two years and 80 credits.

In order to better facilitate the academic and aspirational diversity of our students, RI offers three tracks as follows

- Research Track
- Innovation Track (To be offered in the future, with separate guidelines)
- Incubation Track (To be offered in the future, with separate guidelines)

Each track will have its own set of requirements and passing criteria. Every student must select one of the above 3 tracks to complete their RI requirements. Guidelines for each track will be issued on the required basis, which will be explained every time during the RI workshop and RI course registration.

There will be an RI Governance Council which will consist of representatives from all schools called Research Coordinators, who will be overall responsible for the RI activities from all the schools.

# 2.0 Research Track

The purpose of this track is to enable students with an advanced interest and skill set in research to dig deeper into a particular problem and get "real-world" experience in conducting research. The RI course assessments are aligned with the General Learning Goals that aim to achieve the University's mission of nurturing holistic, socially responsible, and continuously employable individuals.

By the end of the RI course, students will be able to read research papers independently, conduct a literature review, identify research gaps, collect primary/secondary data, perform data analysis, and interpret results from the data analysis. Students will be expected to communicate at least one original research paper to a peer-reviewed research journal (with Scopus indexing/Web of Science indexing) or file an original patent based on their research work.

## 2.1 Objective of Research Track

1. To encourage the students to develop the knowledge, attitudes, and skills to be lifelong, self-directed learners.
2. To help students acquire the knowledge and skills to complete the components of a research project including:
    a) Identification of a challenging problem or an area of research.
    b) Teamwork.
    c) Participation in data collection, analysis, and interpretation.
    d) Application of appropriate methods to arrive at and interpret the results.
    e) Oral dissemination of research information.
    f) Written dissemination of research information.
3. To train students to evaluate the relative effectiveness of alternative solutions for addressing the problem(s).

## 2.2 Process Flow of Research Incubation

### Step I: Coursework

- Students must have completed all GE credits as a prerequisite before starting the RI.
- Students must have completed the SIP course as a prerequisite before starting the RI.
- Students must attend the "Research Methodology" workshop conducted during the first week of course registration. Attending the RI workshop is compulsory. Marks will be allocated based on the performance of the students in a test conducted after the workshop.
- Faculties (RI mentors) should assign courses in Coursera related to the research topic, which must be completed by students within two terms of registration.

### Step II: Research Cluster

The Research Cluster is formed based on the convergence of interest between the faculty and students. The Research cluster/group consists minimum of 2 and a maximum of 4 students from the same school and is assigned to a faculty mentor. However, one faculty mentor can handle only one group. Students should choose faculty from the same school, and the research topics should be relevant to their course.

### Step III: Identification of Research Area

The Research Incubation Program will start with the faculty members submitting their research proposals in their area of interest, which will be circulated to all the students by the respective school RI coordinators. Students along with RI faculty mentors will discuss and finalize the Research Proposal.

### Step IV: Research Framework & Methodology

Once the research problem is identified, the problem will be validated by the research committee, to ensure whether the research problem has already been addressed, or not. To attain this objective, a review of the relevant literature must be carried out. Based on the studies done in the past and the results of those studies, one can know, whether the present research problem has already been addressed or not. Literature review broadens the scope of the research problem, i.e., if the researcher follows the top-down approach, then first

he/she will have to find a broad research area, select a focused research area within that, tabulate all the literature available in that area along with suitable parameters to substantiate the results of the study, and based on this documentation, the research gap will be identified. The research gap would lead to the formulation of the research problem. The research committee will provide advice, guidance, and feedback to the Research Clusters.

**Step V: Credits and Course Duration**

Participation in the Research Incubation Program is mandatory for all students. Students will be graded on their weekly, monthly, milestone, and final presentation performance. Students need to be scored in all the criteria mentioned to obtain 4 credits. RI course should be completed within 4 terms. Term break during RI is subject to consideration and approval of the RI faculty mentor and respective deans of the schools. Course duration and time allocation would be as below:

| Task | Duration | Total no. of hours |
|---|---|---|
| Weekly sessions | 2 Hours/week | 104 |
| Monthly Presentations | 1 Hour/month | 12 |
| Term Presentations (Milestones) | 1 Hour/term | 3 |
| Final Presentation | 1 Hour | 1 |
| **Total no. of hours** | | 120 |

## 2.3 Research Progression

### 2.3.1 Assessment

Specific slots for Research Incubation will be scheduled in the Timetable of the faculty and students post the initial Introduction to Research classroom sessions. The RI slot will be for a minimum of two hours per week per research cluster, which can be both supervised/unsupervised/blend of both and is meant for the faculty to guide and mentor their respective students. Faculty mentors will do a weekly review of the students' timetables, and the progress of the student is assessed as per Annexure I. This will also serve as a time sheet for the continuous evaluation and the grading of each student and must be maintained in LMS. The faculty mentor and Research coordinators should

maintain the Evaluation sheets of the RI as shown in Annexure I. Faculty members should keep an Excel tracker to track the communicated manuscript and should update the RI committee on the changes.

The students will have research progress presentations every month before the faculty mentor and the research committee. The research progress presentation evaluation form is shown in Annexure II. Evaluation of the final presentation will be done by the RI Faculty mentor and RI Coordinators. The final RI presentation evaluation is shown in Annexure III.

RI coordinators will issue a course diary, which is mandatory to maintain in the physical form throughout the course completion. RI dairy would have all the course discussions and weekly sessions details with weekly, monthly, and milestone progress details and assessment records. In case faculties or students are maintaining the course diary online, they should notify the RI coordinator of this deviation and get prior approval.

The students need to complete their research paper before the start of their final Internships. Each Research Cluster will submit its draft paper/progress at the end of each term to the Research Committee for tracking. The Research Committee will do a separate evaluation of the quality of the draft paper/progress, to track the progress of the Research Cluster.

### 2.3.2 Final Review

Every completed research paper should undergo an internal review process by the Research Committee before being communicated to the Journal (Scopus/Web of Science Indexing/UGC Care List 1).

## 2.4 Marks

The weightage of marks for Research Incubation will be distributed across the 5 mandatory submissions: Literature Review, Research Methodology & Data Collection, Data Analysis and Interpretation, first written report, and Final viva Voce. The final marks of each RI cluster will be determined by the Screening Committee depending on the content and publishing potential of the research paper. The student will get a separate certificate for the

RI practice course during their graduation, which will be evaluated based on their milestone achievements. Grades will be given as per the following criteria.

| Session wise | Marks | Weightage in % |
|---|---|---|
| Weekly | 10 | 20 |
| Monthly | 40 | 30 |
| Milestone | 10 | 10 |
| **Final PPT** | **40** | 40 |

## Annexure – 1: Weekly evaluation sheet for RI

| **Weekly Progress Report of RI** | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Title of the Paper:** | | | | | | | | | |
| **Student name/Group:** | | | | | **Faculty name:** | | | | |
| **Meeting No.** | **Date** | **Time** | | **Topic of Discussion** | **Suggestions by Mentor** | **Weekly Grade (10Marks)** | **Student Signature** | **Mentor Signature** | **RI School Coordinator** |
| | | **From** | **To** | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |
| | | | | | | | | | |

**Signed by Faculty Mentor**

**Date:**

**Signed by RI School Coordinator**

**Date:**

RI Faculty need to follow the below assessment metric for weekly grading:

| Mandatory Submissions | Weekly Grading (for 10 Marks) |
|---|---|
| Defining and Structuring the Problem | 2 |
| Work Plan Submission | 2 |
| Analysis and Synthesizing Findings | 2 |
| Development of Recommendations | 2 |
| Final presentation to Faculty | 2 |

## Annexure - II: Form for Monthly Evaluation of RI

Faculty to follow the assessment metric below in the monthly Progress Report or Presentation:

| Student name or Group: | | | |
|---|---|---|---|
| Faculty Mentor: | | | |
| Title of the Paper: | | | |
| **Grading for** | **20 Marks** | | |
| **Presentation Content (if applicable)** | Improvement Needed | Good | Excellent |
| Research Problem Clearly Stated | | | |
| The objectives of the research clearly stated | | | |
| Proper Literature review with a smooth flow | | | |
| Conceptual Framework and theoretical framework | | | |
| **Methodology and Instrumentation (if applicable)** | | | |
| Justification of selected tool | | | |
| Scale development | | | |
| **Visual Aids** | | | |
| Clear catchy slides, not overcrowded | | | |
| Font (size, style, quality) | | | |
| **Performance** | | | |
| Smooth transition from issue to issue | | | |
| Audience contact, eye contact | | | |
| Voice and pacing, Articulation, Engagement, Enthusiasm | | | |
| **General** | | | |
| Ability to answer questions | | | |
| Adherence to the time limit | | | |

**Signed by Faculty Mentor**
**Date:**

**Signed by RI School Coordinator**
**Date:**

# Annexure - III: Form for evaluating Final RI presentations

| S. No | Evaluation Steps and Parameters | Marks & Parameters | Evaluated By | |
|---|---|---|---|---|
| | | | External Faculty | Research Committee |
| 1 | ***Introduction and problem definition*** | ***25 (Marks)*** | | |
| | The problem is clearly stated and inclusion of relevant statistics/information. | 40% | | |
| | Research Paper published in a Journal | 20% | | |
| | Student interest is clearly explained | 20% | | |
| | Relevance of the study | 20% | | |
| 2 | ***Objectives and Literature Review*** | ***20 (Marks)*** | | |
| | Conceptual Framework and theoretical framework | 50% | | |
| | Smooth flow of literature review and adherence to the format (APA, MLA, Harvard) | 50% | | |
| 3 | ***Methodology and Instrumentation*** | ***60 (Marks)*** | | |
| | Justification of selected tool | 25% | | |
| | Scale development | 25% | | |
| | Analysis | 50% | | |
| 4 | ***Discussion and Conclusion*** | ***25 (Marks)*** | | |
| | Content | 40% | | |
| | Synthesis of key points | 40% | | |
| | Scope of future research | 20% | | |
| 5 | ***Dissertation*** | ***20 (Marks)*** | | |
| | Presentation (Slides, Font, PPT Quality) | 20% | | |
| | QA | 60% | | |
| | Adherence to time limit. | 20% | | |

# Annexure - IV: Milestones for RI Course

For the RI practice course, there will be a separate certificate issued along with their degree certificate during convocation. This would be assessed by milestone scores decided by the RI committee. All milestone scores will be sent to Chief of Examination (COE), and the grades in the form of A+, A, B, C, D, and E will be printed on the certificate.

| Milestones | Targets | | 10 Marks |
|---|---|---|---|
| | RI Course registration, Mentor selection | April | Faculty and RI Coordinator |
| 1 | RI workshop | April to June (1st Term Target) | |
| 2 | Problem identification for the RI course | | |
| 3 | Defining Objectives and Problem Statements | | |
| 4 | Literature survey | July to Sep (2nd Term Target) | |
| 5 | Defining Framework, strategy, and Developing an Approach to the Problem | | |
| 6 | Formulating research design | | |
| 7 | Experimental work set up or collection of data | Oct to Dec (3rd Term Target) | |
| 8 | Preparing and analyzing data | | |
| 9 | Preparing and Presenting the Report | | |
| 10 | Conference Paper | Jan to March (Target in Final Term) | |
| 11 | Journal paper submission | | |
| 12 | Final RI Presentation | | |

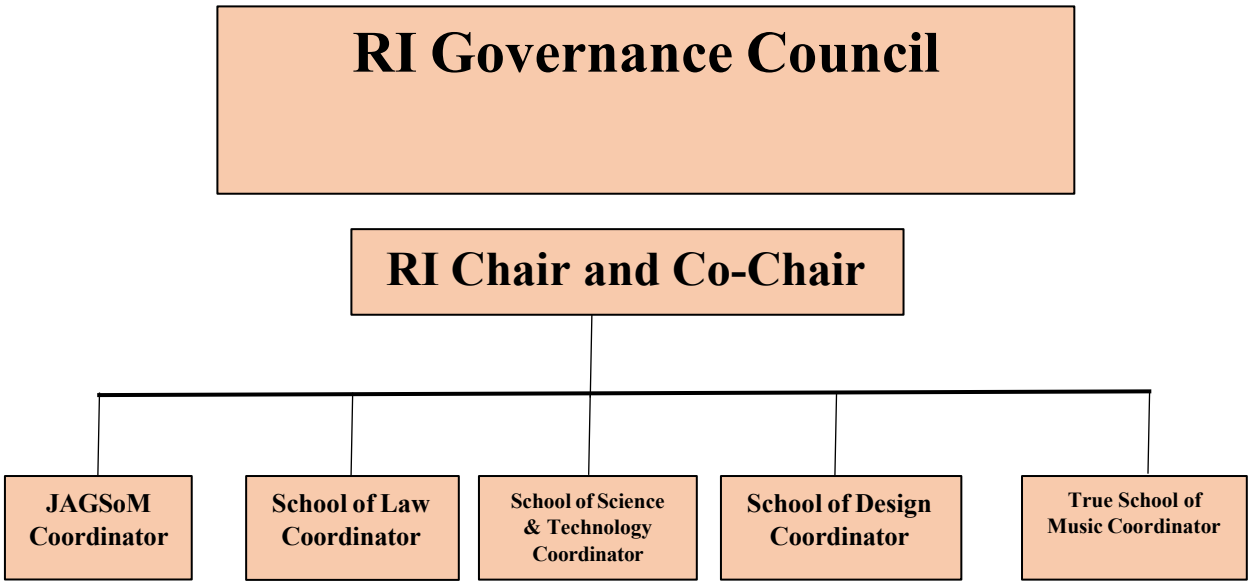
| | |
|---|---|
| A+ | 10 |
| A | 9 |
| B | 8 |
| C | 7 |
| D | 6 |
| E | Below 6 |
| NC | Not Completed |

# 3.0 Operational Framework of RI Tracks

The Organizational Framework for the RI course is given below:

- The Practice Courses Governance Council at Vijaybhoomi University will oversee the execution of the RI for the Undergraduate students at Vijaybhoomi University.
- The Council will be led by the RI Chair and Co-chair.
- The coordinators of each School will work closely with the RI Chair and Co-chair to ensure the delivery and execution of the RI course.

## RI Practice Courses Governance Council




The RI committee will have the responsibility of ensuring that the RI course is delivered to all Undergraduate Students across various schools.

The school RI coordinators will take care of the following steps:

- One month before the start of every term, the school RI coordinator will ask the respective deans of each of the schools to provide the list of students who have completed their credits and all requirements for doing RI.
- Each faculty member needs to submit their research topic(s) to their respective school RI coordinators during the time the RI course is offered.
- Students at each school who are eligible for the RI course will choose the preferred RI topic and faculty only from their respective schools.

- Deans will ensure that the students who have opted for the RI course have completed all the Prerequisites.
- With the approval of deans, the RI school coordinators will finalize the list of RI students from their respective schools.
- Depending on the number of students at the respective schools that have chosen RI, the RI chair and co-chair will work with the respective school RI coordinators to ensure the following:
    a) Sufficient number of Research Proposals (in the correct format) are present and floated for those students who have opted for RI.
    b) For students who wish to convert their RI program to II, a psychometric test should be conducted. Post the psychometric test, the students need to present their ideas to a Jury Committee. In case the students' ideas are not selected by the Jury Committee, the students are to be instructed to continue in the RI program. (Separate guidelines will be issued after considering individual cluster requirements).
- The respective School RI coordinators will ensure that the mandatory weekly meetings between students and faculty are conducted.
- The RI committee will ensure that the research proposals will be translated into publishable research papers.
- It is compulsory for all the students to publish their research work in any of the following formats: Journal articles/Conference papers/Book chapters.
- To get A+/A they have to publish in Scopus listed publications.
- The RI committee will ensure that the Research or Conference papers that are produced as an outcome of the RI Track are communicated to the respective Journals/Conferences only via the faculty's official email ID.
- The RI chair and co-chair along with the respective school RI coordinators will ensure that all the timelines mentioned in the Handbook are adhered to, including the various progress reports and grading.
- The school RI coordinators will highlight any deviations or delays in the process for the various tracks at their respective schools' Faculty Council Meetings. This would help the RI committee to rectify any issues.

# DOCUMENT AMENDMENT RECORD

| S. No | Doc. Ver. No. | Date | Prepared/ Changed By | Reviewed by | Approved By | Brief description of change |
|---|---|---|---|---|---|---|
| 1 | 5.0 | 29/04/2024 | Dr. Krishnakant Bhole | | | New process |
| 2 | 5.1 | 15/05/2024 | Dr. Krishnakant Bhole | Dr. Arindam Ghosh | | New process |
| 3 | 5.2 | 02/06/2024 | Dr. Krishnakant Bhole | President Sir | | New process |
| 4 | 5.3 | 18/07/2024 | Dr. Krishnakant Bhole | Prof. A. Parasuraman | | New process |
| 5 | 5.4 | 30/07/2024 | Dr. Krishnakant Bhole | Dr. Arindam Ghosh | | New process |
| 6 | 5.5 | 19/08/2024 | Dr. Krishnakant Bhole | Prof. A. Parasuraman | | New process |
| 7 | 5.6 | 18/09/2024 | Dr. Krishnakant Bhole | Dr. Arindam Ghosh | Approved by all BOS members | New process |
| 8 | 5.7 | 13/11/2025 | Dr. Ajit Kumar Mohanty | Dr. Arindam Ghosh | 45th Board of Management Meeting held on 14th November | New process |

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